

1 Solution — GIAM 8.4

Problem 1

1.1 Problem

How could the clerk at the Hilbert Hotel accommodate a countable number of new guests?

1.2 Setup

Hilbert's Hotel has rooms indexed by the positive integers $1, 2, 3, \dots$ — one room for each $n \in \mathbb{N}_{>0}$, so $\aleph_0$ rooms in total. The hotel is *full*: every room currently holds exactly one guest, with the bijection

$$g_n \longleftrightarrow \text{room } n \quad (n = 1, 2, 3, \dots).$$

A *countable* set of new guests $h_1, h_2, h_3, \dots$ has just arrived. (In this section's vocabulary, “countable” means countably infinite; the *finite* case was already handled by the textbook's $n \mapsto n + 1$ trick to make room for one new guest, which trivially iterates for any fixed number.) The clerk must find a room for every existing g_n *and* every new h_k , with no two guests sharing.

1.3 The Trick — Double Everyone, Halve the New Arrivals

Have every current guest in room n move to room $2n$, then place new guest h_k in room $2k - 1$.

Guest	Old room	New room
g_n (existing)	n	$2n$
h_k (arriving)	—	$2k - 1$

Table 1.1

A numerical picture of the first eight rooms:

Room	1	2	3	4	5	6	7	8
Before	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8
After	h_1	g_1	h_2	g_2	h_3	g_3	h_4	g_4

Table 1.2

The clerk’s PA announcement is one sentence: “*Every guest in room n , please move to room $2n$.*” Once that move completes, every odd-numbered room is empty, and the arrivals can be checked in: h_1 to room 1, h_2 to room 3, h_3 to room 5, and so on.

1.4 Verification

Three things to check, all easy:

Every existing guest still has a room. The map $n \mapsto 2n$ is a function from $\mathbb{N}_{>0}$ to $\mathbb{N}_{>0}$ — it produces a single, well-defined positive integer for every n .

Every new guest gets a room. The map $k \mapsto 2k - 1$ is also a function from $\mathbb{N}_{>0}$ to $\mathbb{N}_{>0}$.

No collisions. The two image sets are

$$\begin{aligned}\{2n : n \geq 1\} &= \{2, 4, 6, 8, \dots\} && \text{(the evens),} \\ \{2k - 1 : k \geq 1\} &= \{1, 3, 5, 7, \dots\} && \text{(the odds).}\end{aligned}$$

These are disjoint, and together they exhaust $\mathbb{N}_{>0}$. Combined with the obvious injectivity within each group (different n ’s map to different even rooms; different k ’s map to different odd rooms), the assignment is a bijection between the combined guest list and the room list, sending

$$\{g_n\}_{n \geq 1} \cup \{h_k\}_{k \geq 1} \longrightarrow \mathbb{N}_{>0},$$

where the union on the left is *disjoint* (the existing and new guests are different people). Every guest, old and new, ends up in exactly one room. ■

1.5 Remark — Why the "+1" Trick Doesn't Generalize

For *one* new guest, the textbook used $n \mapsto n + 1$: everyone shifts up one room, freeing room 1. For *finitely many*, say m new guests, the same idea works: $n \mapsto n + m$ frees rooms $1, 2, \dots, m$.

But you cannot shift everyone up "by infinity." There is no room number " $n + \aleph_0$ " — the rooms are indexed by natural numbers, not by ordinals. The doubling map $n \mapsto 2n$ is the move that frees *infinitely many* rooms while every output stays a finite natural number; the image just skips every odd integer.

This exploits a defining feature of $\aleph_0$: a proper subset of $\mathbb{N}_{>0}$ (the evens) can be in bijection with $\mathbb{N}_{>0}$ itself — Galileo's paradox, in essence. Finite sets cannot do this; infinite ones can.

1.6 Remark — Other Bijections That Work

Doubling-and-halving is the cleanest scheme, but it isn't the only choice. *Any* pair of injections $\alpha, \beta : \mathbb{N}_{>0} \hookrightarrow \mathbb{N}_{>0}$ with disjoint images whose union covers $\mathbb{N}_{>0}$ will work:

Existing g_n goes to $\alpha(n)$	New h_k goes to $\beta(k)$	Comment
$2n$ (evens)	$2k - 1$ (odds)	The standard move.
$2n - 1$ (odds)	$2k$ (evens)	Symmetric; just swaps the two groups.
$n + k_0$ for some fixed k_0	$1, 2, \dots, k_0$	Fails — only k_0 rooms freed, not infinitely many.

Table 1.3

The last row is the failure mode the previous remark warned about: a finite shift creates only finitely many vacancies, and the $k_0 + 1$ -st new guest has nowhere to go.

1.7 Remark — Going Further, Countably Many Buses

The next layer of difficulty: a *countable number of countable busloads* arrives — bus i contains guests $h_{i,1}, h_{i,2}, h_{i,3}, \dots$ for each $i = 1, 2, \dots$. The clerk now needs to accommodate $\aleph_0 \cdot \aleph_0$ new guests.

The same doubling idea scales via a *pairing function*. One clean choice uses prime-power encoding: send g_n to room 2^n and bus passenger $h_{i,j}$ to room p_i^j , where $p_1, p_2, \dots$ are the *odd* primes ($3, 5, 7, \dots$). These rooms are all distinct by the fundamental theorem of arithmetic, so the assignment is an injection — every guest gets a unique room, though now a lot of rooms (those whose prime factorization mixes 2 with an odd prime, like 6 or 12) sit empty.

Alternatively, the *Cantor pairing function*

$$\pi(i, j) = \frac{(i+j)(i+j+1)}{2} + j$$

gives an explicit *bijection* $\mathbb{N}_{>0} \times \mathbb{N}_{>0} \rightarrow \mathbb{N}_{>0}$, which combined with $g_n \mapsto 2n$ (still occupying the evens) lets the bus passengers fill the odd rooms with no waste.

The moral: *every countable union of countable sets is countable*, so the hotel can always absorb such crowds. The clerk only runs out of rooms when a genuinely *uncountable* convention arrives — e.g., one guest for each real number — which is the content of Cantor’s theorem in this section’s language.

1.8 Remark — Reading This Through the Lens of Dominance

This solution is, in Section 8.4 terms, exhibiting an *injection* from “all guests, existing plus new” into $\mathbb{N}_{>0}$ (the rooms). That is,

$$|\{g_n\}_{n \geq 1} \cup \{h_k\}_{k \geq 1}| \leq |\mathbb{N}_{>0}| = \aleph_0,$$

with the union disjoint. The reverse direction $\aleph_0 \leq |\text{all guests}|$ is trivial (every existing guest is already in the combined set, so there are at least $\aleph_0$ of them). By the Cantor-Bernstein-Schröder theorem,

$$|\{g_n\}_{n \geq 1} \cup \{h_k\}_{k \geq 1}| = \aleph_0,$$

i.e. $\aleph_0 + \aleph_0 = \aleph_0$. The Hilbert Hotel is a concrete, narrative form of this cardinal-arithmetic identity.